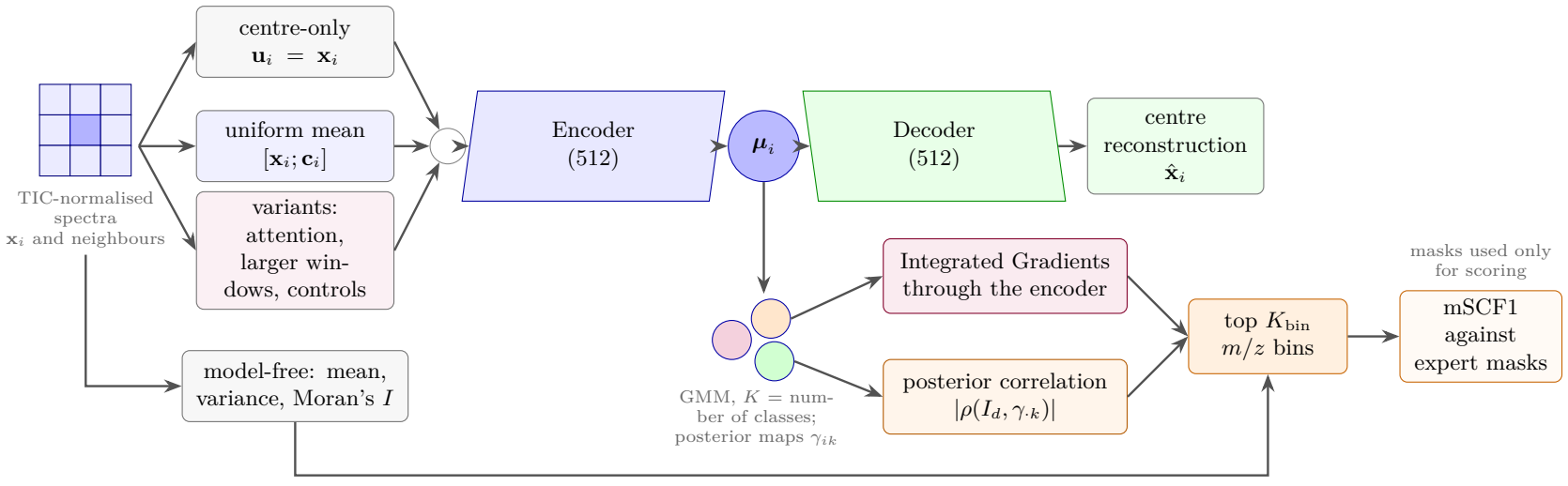

centre-only
$\mathbf{u}_i = \mathbf{x}_i$
uniform mean
$[\mathbf{x}_i; \mathbf{c}_i]$
variants:
attention,
larger win-
dows, controls
TIC-normalised
spectra
$\mathbf{x}_i$ and neighbours
Encoder
(512)
$\boldsymbol{\mu}_i$
Decoder
(512)
centre
reconstruction
$\hat{\mathbf{x}}_i$
model-free: mean,
variance, Moran's I
GMM, K = num-
ber of classes;
posterior maps γ_{ik}
Integrated Gradients
through the encoder
posterior correlation
$|\rho(I_d, \gamma_{\cdot k})|$
top K_{bin}
m/z bins
masks used only
for scoring
mSCF1
against
expert masks